Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A particle executes SHM with an amplitude $8\ \text{cm}$ and a frequency $10\ \text{s}^{-1}$. Assuming the particle to be at a displacement $4\ \text{cm}$ initially, in the positive direction, determine its displacement equation and the maximum velocity and acceleration.

(A) $3200\ \pi^2\text{s}$

(B) $3800\ \pi^2\text{s}$

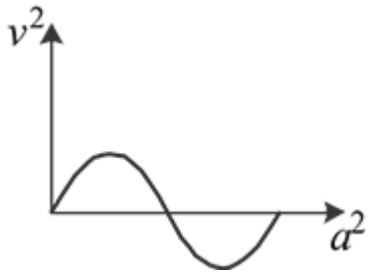
(C) $7200\ \pi^2\text{s}$

(D) $2000\ \pi^2\text{s}$

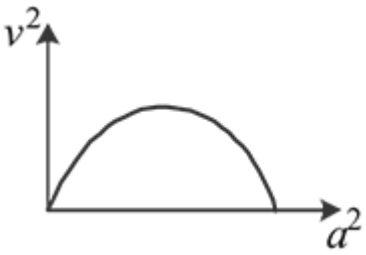
Q.2

A graph of the square of the velocity against the square of the acceleration of a given simple harmonic motion is

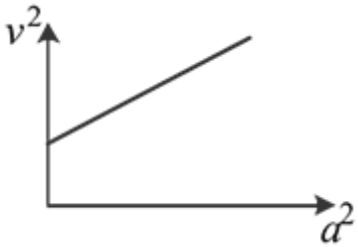
(A)



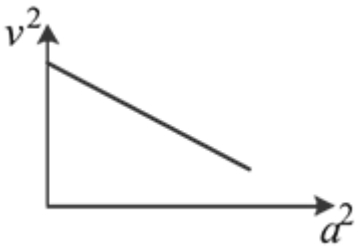
(B)



(C)



(D)



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Q.3

For a particle moving according to the equation $x = a\cos\pi t$, the displacement in 3 s is

(A) 0

(B) $0.5\mathrm{a}$

(C) $1.5\mathrm{a}$

(D) $2\mathrm{a}$

Q.4

A body oscillates with SHM according to the equation (in SI units), $x = 5\cos\left(2\pi + \frac{\pi}{4}\right)$. Its instantaneous displacement at $t = 1$ second is

(A) $\frac{\sqrt{2}}{5}$ m

(B) $\frac{1}{\sqrt{3}}$ m

(C) $\frac{5}{\sqrt{2}}$ m

(D) $\frac{1}{2}$ m

Q.5

A point mass oscillates along the x -axis according to the law $x = x_0\cos(\omega t - \pi/4)$. If the acceleration of the particle is written as $\mathrm{a} = \mathrm{A}\cos(\omega\mathrm{t} + \delta)$, then

(A) $\mathrm{A} = \mathrm{x}_0\omega^2, \delta = 3\pi/4$

(B) $A = x_0, \delta = -\pi/4$

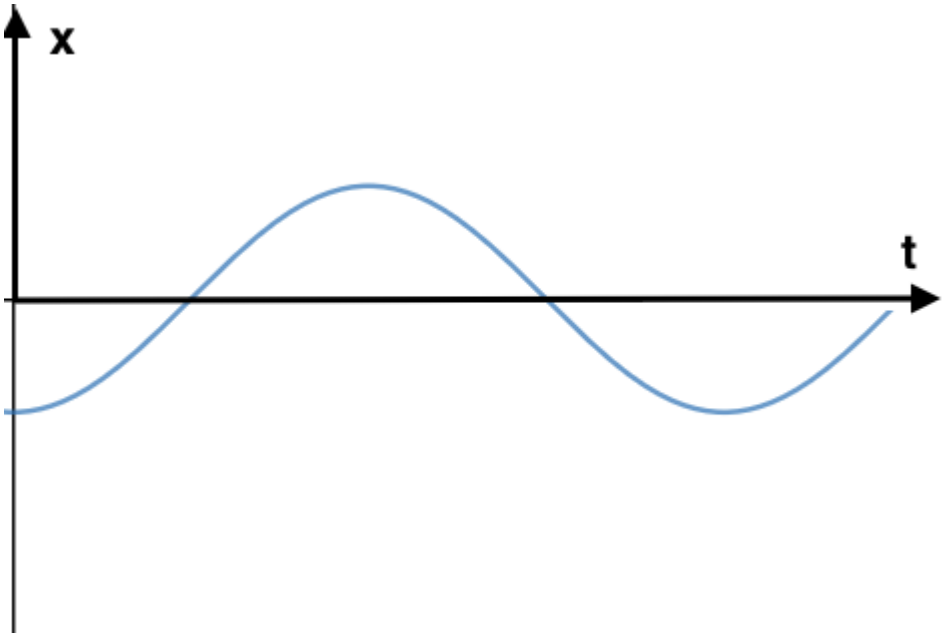
(C) $\mathrm{A} = \mathrm{x}_0\omega^2, \delta = \pi/4$

(D) $A = x_0\omega^2, \delta = -\pi/4$

Q.6

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The displacement vs time of a particle executing SHM is shown in figure. The initial phase ϕ is



(A) $-\pi < \phi < -\frac{\pi}{2}$

(B) $\pi < \phi < \frac{3\pi}{2}$

(C) $-\frac{3\pi}{2} < \phi < -\pi$

(D) $\frac{\pi}{2} < \phi < \pi$

Q.7

The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is $16\ \mathrm{cm/sec}$. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}\ \mathrm{cm/s}$, will be

(A) $2\sqrt{3}\ \mathrm{cm}$

(B) $\sqrt{3}\ \mathrm{cm}$

(C) $1\ \mathrm{cm}$

(D) $2\ \mathrm{cm}$

Q.8

Two simple harmonic motions are represented by the equations $\mathrm{y_1} = 0.1 \sin\left(100\pi \mathrm{t} + \frac{\pi}{3}\right)$ and $\mathrm{y_2} = 0.1 \cos \pi \mathrm{t}$.
The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is

(A) $\frac{\pi}{3}$

(B) $\frac{-\pi}{6}$

(C) $\frac{\pi}{6}$

(D) $\frac{-\pi}{3}$

Q.9

Two particles are oscillating along two close parallel straight lines side by side, with the same frequency and amplitudes. They pass each other, moving in opposite directions when their displacement is half of the amplitude.

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The mean positions of the two particles lie on a straight line perpendicular to the paths of the two particles. The phase difference is

(A) 0

(B) $2\pi/3$

(C) π

(D) $\pi/6$

Q.10

Find the resultant amplitude of the following simple harmonic equations:

$x_1 = 5\sin \omega t$

$x_2 = 5\sin\omega t + 53°$

$x_3 = -10\cos\omega t$

(A) 5

(B) 10

(C) 15

(D) 20

Q.11

A particle performs simple harmonic motion with amplitude A. It's speed is tripled at the instant that it is at a distance 2 A/3 from the equilibrium position. The new amplitude of the motion is:

(A) $\frac{A}{3}\sqrt{41}$

(B) 3 A

(C) A B

(D) $\frac{7}{3}$ A

Q.12

A particle executing SHM has velocities u and and acceleration a and b in two of its positions. Find the distance between these two positions.

(A) $\frac{u^2-v^2}{a+b}$

(B) $\frac{v^2+u^2}{a-b}$

(C) $\frac{v^2+u^2}{a+b}$

(D) $\frac{v^2-u^2}{a-b}$

Q.13

A mass m_1 connected to a horizontal spring performs S.H.M. with amplitude A. While mass m_1 is passing through

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mean position another mass m_2 is placed on it so that both the masses move together with amplitude A_1. The ratio of $\frac{A_1}{A}$ is $(m_2 < m_1)$

(A) $\left[\frac{m_1}{m_1+m_2}\right]^{\frac{1}{2}}$

(B) $\left[\frac{m_1+m_2}{m_1}\right]^{\frac{1}{2}}$

(C) $\left[\frac{m_2}{m_1+m_2}\right]^{\frac{1}{2}}$

(D) $\left[\frac{m_1+m_2}{m_2}\right]^{\frac{1}{2}}$

Q.14

A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency ω. The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time

(A) at the mean position of the platform

(B) for an amplitude of $\frac{g}{\omega^2}$

(C) for an amplitude of $\frac{g^2}{\omega^2}$

(D) at the bottom most position of the platform

Q.15

A simple harmonic oscillator of angular frequency $\mathrm{rad\,s^{-1}}$ is acted upon by an external force $F = \sin t$ N. If the oscillator is at rest in its equilibrium position at $t = 0$, its position at later times is given by:

(A) $\sin t + \frac{1}{2}\cos 2t$

(B) $\cos t - \frac{1}{2}\sin 2t$

(C) $\sin t - \frac{1}{2}\sin 2t$

(D) $\sin t + \frac{1}{2}\sin 2t$

Q.16

Two particles execute SHM on same straight line with same mean position, same time period 6 second and same amplitude 5 cm. Both the particles start SHM from their mean position (in same direction) with a time gap of 1 second. The maximum separation between the particles during their motion is

(A) 2 cm

(B) 3 cm

(C) 4 cm

(D) 5 cm

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Q.17

A particle is executing simple harmonic motion with amplitude A. When the ratio of its kinetic energy to the potential energy is $\frac{1}{4}$, its displacement from its mean position is

(A) $\frac{2}{\sqrt{5}}\ \mathrm{A}$

(B) $\frac{\sqrt{3}}{2}\ \mathrm{A}$

(C) $\frac{3}{4}\ \mathrm{A}$

(D) $\frac{1}{4}\ \mathrm{A}$

Q.18

A mass of $4\ \mathrm{kg}$ suspended from a spring of force constant $800\ \mathrm{N\ m^{-1}}$ executes simple harmonic oscillations. If the total energy of the oscillator is $4\ \mathrm{J}$, the maximum acceleration (in $\mathrm{ms^{-2}}$) of the mass is

(A) 5

(B) 15

(C) 45

(D) 20

Q.19

If the maximum velocity and acceleration of a particle executing SHM are equal in magnitude, the time period will be

(A) $1.57\mathrm{sec}$

(B) $3.14\mathrm{sec}$

(C) $6.28\mathrm{sec}$

(D) $12.56\mathrm{sec}$

Q.20

A particle executes simple harmonic motion between $\mathrm{x} = -\mathrm{A}$ and $\mathrm{x} = +\mathrm{A}$. The time taken by it to go from O to $\mathrm{A}/2$ is $\mathrm{T_1}$ and to go from $\mathrm{A}/2$ to A is $\mathrm{T_2}$. Then

(A) $\mathrm{T_1} < \mathrm{T_2}$

(B) $\mathrm{T_1} > \mathrm{T_2}$

(C) $\mathrm{T_1} = \mathrm{T_2}$

(D) $\mathrm{T_1} = 2\ \mathrm{T_2}$

Answers & Solutions

Q.1 Answer:

$3200\ \pi^2 \mathrm{s}$

Solution:

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Given $a = 8$ cm and $\omega = 2\pi n = 20\pi \text{rad/s}$

Let the phase constant be ϕ.

The displacement equation can be written as $x = 8\sin(20\pi t + \phi)$

Given, at $t = 0; x = 4$ cm, therefore

$4 = 8\sin(20\pi(0) + \phi) \quad \Rightarrow \sin(\phi) = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{6}$

$\therefore$ Displacement equation $x = 8\sin\left(20\pi t + \frac{\pi}{6}\right)$

Differentiating the above equation w.r.t. time 't'

$\frac{dx}{dt} = v = 160\pi\cos\left(20\pi t + \frac{\pi}{6}\right)$

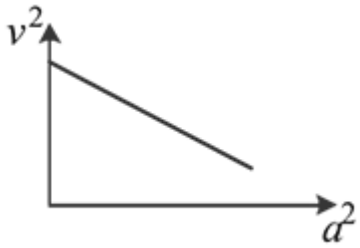
$\therefore \quad v_{\max} = \pm 160\pi \text{cm/s} \quad [\text{ when } \cos(20\pi t + \pi 6) = \pm 1$

Differentiating once again w.r.t. time t.

$\frac{d^2x}{dt^2} = a = -3200\pi^2\sin\left(20\pi t + \frac{\pi}{6}\right)$

$a_{\max} = \pm 3200\pi^2$

Q.2 Answer:



Solution:

$v^2 = \omega^2\left(A^2 - x^2\right)$

... (i) and $a^2 = \left(\omega^2 x^2\right) = \omega^4 x^2$ ... (ii)

From above equations, we have $v^2 = -\frac{a^2}{\omega^2} + \omega^2 A^2 \Rightarrow y = mx + c$

It represents straight line with negative slope.

Q.3 Answer:

2a

Solution:

Initially $t = 0\ x = a\cos\pi t = -a\cos 0^\circ = a$

Finally at $t = 3\ x = a\cos 3\pi = -a$

Total displacement $= 2\text{a}$

Q.4 Answer:

$\frac{5}{\sqrt{2}}$ m

Solution:

Given $t = 1$ s $\therefore x = 5\cos\left(2\pi + \frac{\pi}{4}\right) = 5\cos\frac{\pi}{4} = \frac{5}{\sqrt{2}}$

i.e., displacement at $t = 1$ s is $\frac{5}{\sqrt{2}}$ m

Q.5 Answer:

$\text{A} = \text{x}_0\omega^2, \delta = 3\pi/4$

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Solution:

Here, displacement $x = x_0 \cos(\omega t - \pi/4)$
Velocity, $\mathrm{v} = \frac{\mathrm{dx}}{\mathrm{dt}} = -\mathrm{x}_0\omega \sin(\omega \mathrm{t} - \pi/4)$
Acceleration, $\mathrm{a} = \frac{\mathrm{dv}}{\mathrm{dt}} = -\mathrm{x}_0\omega^2 \cos(\omega \mathrm{t} - \pi/4)$
or $\mathrm{a} = \mathrm{x}_0\omega^2 \cos(\pi + \omega \mathrm{t} - \pi/4)$
or $\mathrm{a} = \mathrm{x}_0\omega^2 \cos(\omega \mathrm{t} - 3\pi/4) \ldots (1)$
Also given, $a = A\cos(\omega t + \delta) \ldots (2)$
Comparing (1) and (2), we get $\mathrm{A} = \mathrm{x}_0\omega^2$ **and** $\delta = \frac{3\pi}{4}$

Q.6 Answer:

$-\pi < \phi < -\frac{\pi}{2}$

Solution:

For $x = (-\mathrm{A})$**, we have** $-\mathrm{A} = \mathrm{A}\sin(\omega \times 0 + \phi_0)$ **or** $\phi_0 = -\frac{\pi}{2}$
So for $x < (-A), \phi_0 < (-\pi/2)$.

Q.7 Answer:

2 cm

Solution:

At mean position velocity is maximum
i.e., $v_{\max} = \omega a \Rightarrow \omega = \frac{v_{\max}}{a} = \frac{16}{4} = 4$
$\therefore v = \omega\sqrt{a^2 - y^2} \Rightarrow 8\sqrt{3} = 4\sqrt{4^2 - y^2}$
$\Rightarrow 192 = 16\left(16 - y^2\right) \Rightarrow 12 = 16 - y^2$
$\Rightarrow y = 2$ cm

Q.8 Answer:

$\frac{-\pi}{6}$

Solution:

$$\mathrm{v}_1 = \frac{\mathrm{dy}_1}{\mathrm{dt}} = 0.1 \times 100\pi \cos\left(100\pi \mathrm{t} + \frac{\pi}{3}\right)$$

$$v_2 = \frac{dy_2}{dt} = -0.1\pi \sin \pi t$$

$$= 0.1\pi \cos\left(\pi t + \frac{\pi}{2}\right)$$

$$\therefore \text{ Phase diff. } = \phi_1 - \phi_2 = \frac{\pi}{3} - \frac{\pi}{2} = \frac{2\pi - 3\pi}{6} = -\frac{\pi}{6}$$

Q.9 Answer:

$2\pi/3$

Solution:

Equation of S.H.M. is given by $\mathrm{x} = \mathrm{A}\sin(\omega \mathrm{t} + \delta)$
$(\omega \mathrm{t} + \delta)$ **is called phase. When** $\mathrm{x} = \frac{\mathrm{A}}{2}$**, then** $\sin(\omega \mathrm{t} + \delta) = \frac{1}{2}$

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$\Rightarrow \omega t + \delta = \frac{\pi}{6}$

or $\phi_1 = \frac{\pi}{6}$

For second particle,

$\phi_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$\therefore \phi = \phi_2 - \phi_1 = \frac{4\pi}{6} = \frac{2\pi}{3}$

Q.10 Answer:

10

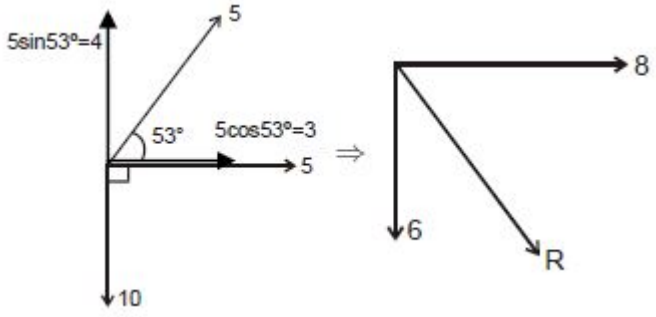
Solution:

$x_1 = 5\sin \omega t$

$x_2 = 5\sin\omega t + 53°$

$x_3 = -10\cos\omega t$

we can write $x_3 = 10\sin\omega t - 90°$ **Finding the resultant amplitude by vector notation.**



Resultant Amplitude $|R| = \sqrt{8^2 + 6^2} = 10.$

Q.11 Answer:

$\frac{7}{3}$ A

Solution:

The velocity of a particle executing SHM at any instant is defined as the time rate of change of its displacement at that instant.

$v = \omega\sqrt{A^2 - x^2}$

Where ω **is angular frequency,** A **is amplitude and** x **are displacements of a particle.**

Suppose that the new amplitude of the modules be A.

The initial velocity of a perticle performs SHM.

$v^2 = w^2\left[A^2 - \left(\frac{2A}{3}\right)^2\right]$

Where A, is initial amplitude and ω **is angular frequency.**

Final velocity.

$\left(3v^2\right) = \omega^2\left[A^2 - \left(\frac{2A}{3}\right)^2\right]$

From Eqs. (i) and (ii), we get

$\frac{1}{9} = \frac{A^2 - \frac{4A^2}{9}}{A^2 - \frac{4A^2}{9}}$

$A' = \frac{7A}{3}$

Q.12 Answer:

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$\frac{u^2-v^2}{a+b}$

Solution:

Let x_1 and x_2 be the distances of the two positions from centre. Then with usual notations:

$$u^2 = \omega^2 \left(A^2 - x_1^2 \right) ...(i)$$

$$v^2 = \omega^2 \left(A^2 - x_2^2 \right) ...(ii)$$

$$a = \omega^2 x_1 ...(iii)$$

$$b = \omega^2 x_2 ...(iv)$$

Subtracting Eq (ii) from (i),

$$u^2 - v^2 = \omega^2 \left(x_2^2 - x_1^2 \right) ...(v)$$

Adding (iii) and (iv),

$$a + b = \omega^2 \left(x_1 + x_2 \right) ...(vi)$$

Dividing Eq (v) by eq. (vi)

$$\frac{u^2-v^2}{a+b} = x_2 - x_1$$

Q.13 Answer:

$$\frac{m_1}{m_1 + m_2}^{\frac{1}{2}}$$

Solution:

P.E of the oscillating mass is given by

$$E = \frac{1}{2} m \omega x^2 = \frac{1}{2} k x^2$$

where $k = m\omega^2$

Let $k = m\omega^2$

be the velocity at mean position. When mass M_2

is attached $v_2 = \frac{m_1 v_1}{m_1 + m_2}$

If A_1

be the new amplitude $\frac{1}{2} k A_1^2 = \frac{1}{2} \left(m_1 + m_2 \right) v_2^2$

$$\frac{1}{2} \left(m_1 + m_2 \right) \frac{m_1^2 v_1^2}{\left(m_1 + m_2 \right)^2}$$

$$= \frac{1}{2} m_1 v_1^2 \times \frac{m_1}{m_1 + m_2}$$

$$= \frac{1}{2} k A^2 \times \frac{m_1}{m_1 + m_2}$$

$$\Rightarrow \frac{A_1}{A} = \sqrt{\frac{m_1}{m_1 + m_2}}$$

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Q.14 Answer:

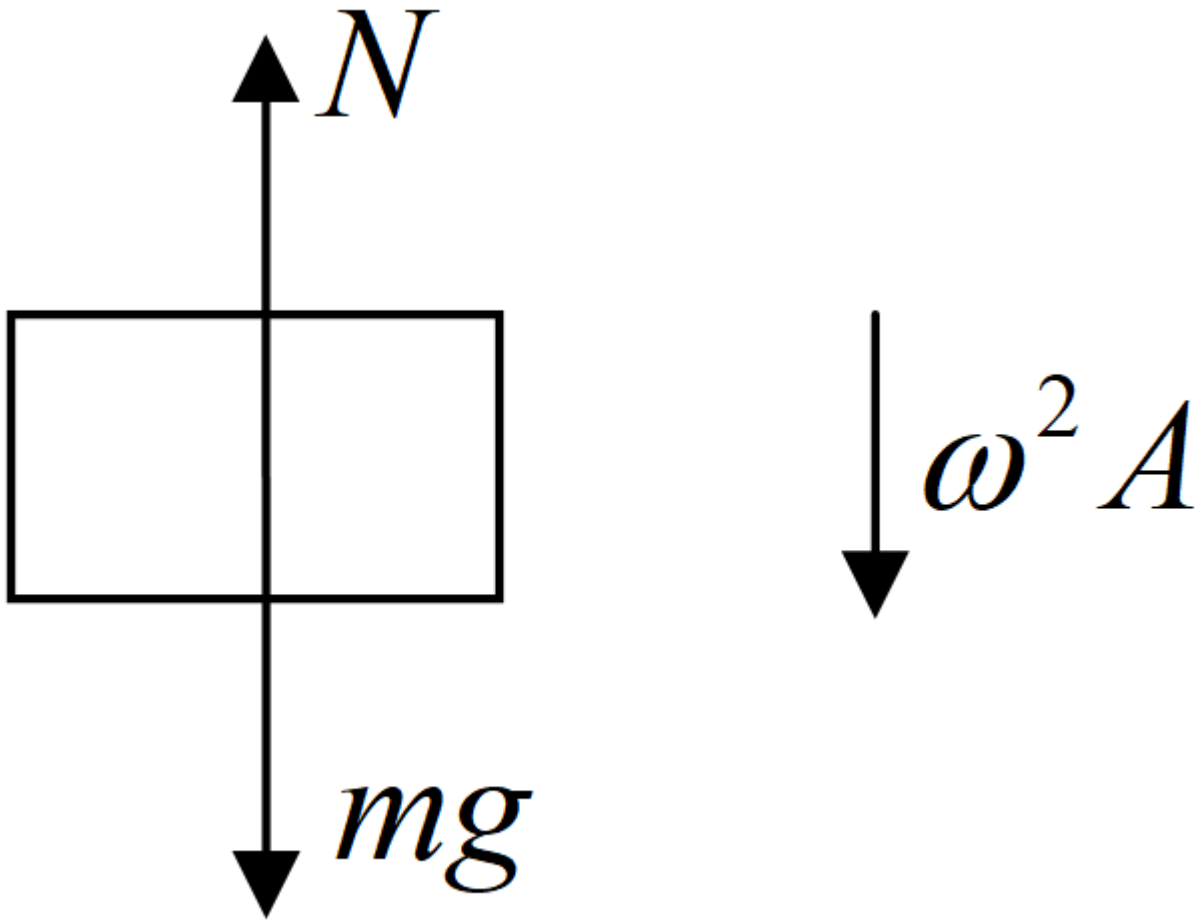
for an amplitude of $\frac{g}{\omega^2}$

Solution:

The maximum acceleration $\omega^2 A$ of the coin and platform increases as the amplitude increases. Here ω is the angular frequency and A is the amplitude of oscillation.

The coin will leave contact with the platform for the first time at the top extreme position as its acceleration will be maximum in the downward direction at this position.

Let us draw the FBD of coin at the top extreme position:



Applying Newton's second law, we get, $mg - N = m\omega^2 A$

If coin loses contact, then normal reaction becomes zero. Putting $N = 0$**, we get** $A = \frac{g}{\omega^2}$

Q.15 Answer:

$\sin t - \frac{1}{2}\sin 2t$

Solution:

As we know, $F = ma \Rightarrow a \propto F$ **or,** $a \propto \sin$

$\Rightarrow \frac{dv}{dt} \propto \sin t \Rightarrow \int_0^0 dV \propto \int_0^t \sin t dt$

$\mathrm{V} \propto -\cos t + 1$

$\int_0^x dx = \int_0^t (-\cos t + 1) dt$

$x = \sin t - \frac{1}{2}\sin 2t$

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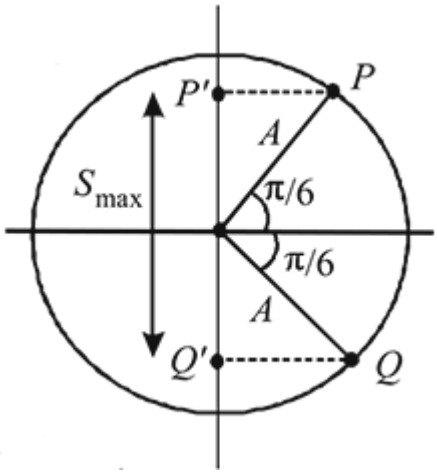
Q.16 Answer:

5 cm

Solution:

Figure shows the mapping of the SHMs with circular motions having phase difference

or $S_{max} = 2 \times 5 \times \frac{1}{2} = 5$ cm



Q.17 Answer:

$\frac{2}{\sqrt{5}}$ A

Solution:

$\therefore \frac{\frac{1}{2}m\omega^2\left(A^2-x^2\right)}{\frac{1}{2}m\omega^2x^2} = \frac{1}{4} \Rightarrow \frac{A^2-x^2}{x^2} = \frac{1}{4}$

$4A^2 - 4x^2 = x^2 \Rightarrow x^2 = \frac{4}{5}A^2 \Rightarrow x = \frac{2}{\sqrt{5}}A.$

Q.18 Answer:

20

Solution:

No solution available

Q.19 Answer:

6.28sec

Solution:

No solution available

Q.20 Answer:

$T_1 < T_2$

Solution:

The equation of SHM, when it starts swinging from the mean position, is $y = A\sin\omega t$.

At mean position, $y = o, t = O \quad \ldots..(1)$

$\frac{A}{2} = A\sin\omega t_2$ **or** $\sin\omega t_2 = \frac{1}{2} = \sin\frac{\pi}{6}$

$\omega t_2 = \frac{\pi}{6}$ **or** $t_2 = \frac{\pi}{6\omega} \quad \ldots..(2)$

At $y = A$ **time for travel from** o **to** $A = t_3$

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$A = A \sin \omega t_3$

or $\sin \omega t_3 = 1 = \sin \frac{\pi}{2}$ **or** $t_3 = \frac{\pi}{2\omega}$ .....(3)

$T_1 = t_2 - t_1 = \frac{\pi}{6\omega}$ .....(4)

$T_2 = t_3 - t_2 = \frac{\pi}{2\omega} - \frac{\pi}{6\omega} = \frac{\pi}{3\omega}$ .....(5)

$\frac{T_1}{T_2} = \frac{\frac{\pi}{6\omega}}{\frac{\pi}{3\omega}} = \frac{3\omega}{6\omega} = \frac{1}{2}$ **or** $\frac{T_1}{T_2} = \frac{1}{2}$

$T_1 < T_2$

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